

Sreya Sarkar

Curriculum Vitae

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[LinkedIn](#) / [GitHub](#)

Education

- 2023 - May 2026 **Ph.D., Statistics**, The University of Iowa (UI), Iowa City, USA.
(Expected) GPA: 3.9/4
Advisor: [Dr. Sanvesh Srivastava](#)
- 2021 - 2023 **M.S., Statistics**, The University of Iowa, Iowa City, USA.
GPA: 3.8/4
Advisor: [Dr. Sanvesh Srivastava](#)
- 2019 - 2021 **M.Sc., Statistics**, Indian Institute of Technology (IIT), Kanpur, India.
GPA: 8.5/10
Advisor: [Dr. Minerva Mukhopadhyay](#)
- 2016 - 2019 **B.Sc. (Honours), Statistics**, St. Xavier's College (SXC), Kolkata, India.
GPA: 8.6/10
Advisor: [Dr. Surupa Chakraborty](#)

Research Interests

High-dimensional repeated-measures data, Multimodal data, Tensor Regression, Bayesian Statistics, Computational Statistics.

Experience

- May 2023–
Ongoing **Graduate Research Assistant**, *DEPARTMENT OF STATISTICS, THE UNIVERSITY OF IOWA*,
Supervisor: [Dr. Sanvesh Srivastava](#).
- Developed a computationally efficient Bayesian methodology for prediction and fixed effects selection with mixed-effects models in high-dimensional repeated-measures data with array-variate covariates.
 - Leveraged random projections to handle high-dimensional random effects covariance matrices and established theoretical guarantees of the associated Bayes risk of the proposed method.
 - Demonstrated the advantages of the proposed approach over existing penalized quasi-likelihood methods through extensive simulation studies and real-data analyses in biomedical and behavioral applications.
- Aug 2021–
Ongoing **Graduate Teaching Assistant**, *DEPARTMENT OF STATISTICS, THE UNIVERSITY OF IOWA*.
- Teaching Assistant for courses including Bayesian Statistics (STAT 4520), Statistical Learning (STAT 4540), Probabilistic Statistical Learning (DATA 4750), Probability and Statistics for Engineering (STAT 2020), Statistics for Business (STAT 1030).
- Summer 2024 **Research Assistant**, *DEPARTMENT OF GASTROENTEROLOGY AND HEPATOLOGY, WEILL CORNELL MEDICINE*,
Supervisor: [Dr. Himel Mallick](#).
- Developed a Bayesian cooperative learning method for multimodal integration, utilizing intermediate fusion and Bayesian regularization for continuous, binary, and survival outcomes.
 - Developed a meta-network analysis method to integrate multiple microbiome studies, using concepts from multi-study machine learning and differential network analysis.
 - Conducted comprehensive benchmarking of these methods using realistic synthetic data and applied them to real-world microbiome and multi-omics datasets, including pregnancy outcomes and cancer immunotherapy response.

Publications/Preprints

- In Submission

- 2025+ **Sarkar, S.**, Khare, K., Srivastava, S., Bayesian Compressed Mixed-Effects Models, [arXiv](#).
- 2025+ Roy, S.*, **Sarkar, S.***, Paul E., Basak P., Yi N., Mallick, H., Bayesian Cooperative Learning for Multimodal Integration, [bioRxiv](#) * These authors contributed equally.

- In Preparation

- 2025+ **Sarkar, S.**, Khare, K., Srivastava, S., Bayesian Array-variate Compressed Mixed Models.
- 2025+ **Sarkar, S.**, Roy, S., Zhang, X., Rahanavard, A., Mallick, H., Multi-study Meta-Analysis of Differential Microbiome Networks.
- 2025+ Roy, S., **Sarkar, S.**, Banerjee, C., Mallick, H., Conformalized Multiview Learning.

R Software

- Developer
- o [cme](#), Bayesian Compressed Mixed-Effects Models.
 - o [BayesCOOP](#), Bayesian Cooperative Learning.
 - o [muffinette](#), Multi-study Meta-network Analysis.
 - o [Coracle](#), Conformalized Multiview Learning.

Conferences & Workshops

- Aug 2025 **Invited Talk**, on “*Bayesian Cooperative Learning for Multimodal Integration*”, Joint Statistical Meetings (JSM) 2025, Nashville, USA.
- Aug 2025 **Contributed Talk**, on “*Bayesian Compressed Mixed-Effects Models*”, JSM 2025, Nashville, USA.
- Apr 2025 **Lightning Talk**, on “*Bayesian Compressed Mixed-Effects Models*”, Bayesian Young Statisticians Meeting (BAYSM) 2025, j-ISBA, Virtual.
- Dec 2024 **Poster Presentation**, on “*Bayesian Compressed Mixed-Effects Models*”, International Indian Statistical Association (IISA) Conference 2024, Kochi, India.
- Dec 2024 **Omics Data Science Workshop**, IISA Conference 2024, Kochi, India.

Awards

- Aug 2025 **UI Graduate Student Senate Travel Award**, - JSM 2025, Nashville, USA.
- Dec 2024 **NSF Travel Award**, - IISA Conference 2024, Kochi, India.
- Jul 2019 **Fr. Lafont Scholarship for Academic Performance**, - SXC Kolkata, India.
- July 2016 - **Department of Science and Technology, INSPIRE Scholarship**, - Government of India.
- July 2021

Professional Services

- Sep 2024-Present **Reviewer**, *Electronic Journal of Statistics, Sankhya B, STAT*.
- Aug 2025 **Session Chair**, Invited Session on “*Advances in Foundation Models and LLMs for Biomedical Data Science*”, **JSM 2025**, Nashville, USA.
- Dec 2024 **Session Chair**, Panel Discussion on “*Breaking Down Barriers and Cultivating a Culture of Collaboration and Leadership in Statistical Sciences: Enhancing Synergy for Collective Growth*”, **IISA Conference 2024**, Kochi, India.
- Apr 2024 - Present **Treasurer**, *Statistics Student Organization*, Department of Statistics and Actuarial Science, The University of Iowa.